

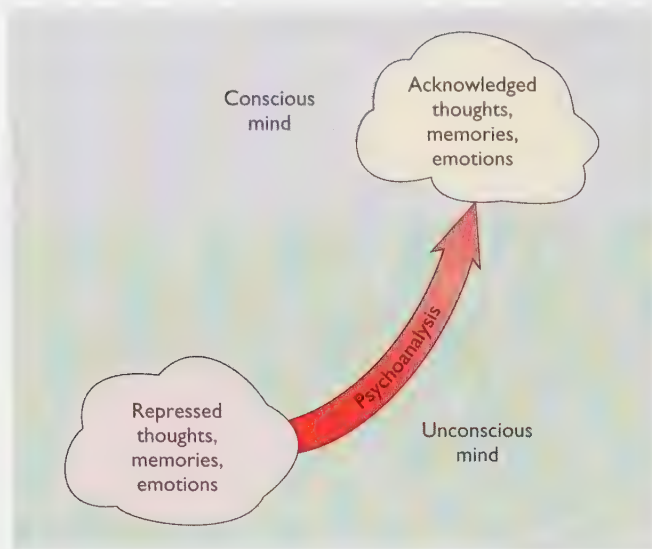


FIGURE 13.10 Freud believed that, through psychoanalysis, parts of the unconscious could be brought into the conscious mind, where a person could deal with them.

As Freud listened to his patients, however, he became convinced that the traumatic events they recalled were not sufficient to account for their abnormal behavior. Some patients reacted strongly to events that others took in stride. Why? At first, in the early 1890s, Freud attributed their overreactions to sexual difficulties (Macmillan, 1997). People with excessive anxiety, he said, were suffering from a lack of sexual gratification, and they could solve their problems by increased sexual activity. (For a while, Freud claimed that this recommendation cured their problems, but he soon abandoned it, and since then no one has found it to be effective.) People suffering from nervous exhaustion, he said, were suffering from the results of masturbation. His “evidence” for this idea was that all of his patients suffering from nervous exhaustion had masturbated (!). You can see the problem: Freud was practicing long before the Kinsey report had been published, and no one knew yet how common masturbation was. Freud boldly asserted that normal, mentally healthy people do not masturbate. Now, suppose you were in his audience. Would you raise your hand and say, “Wait a minute; I think I’m mentally normal, and I masturbate. . .”? Probably not, and neither did anybody else, so Freud concluded that he was right.

A few years later, however, Freud abandoned these hypotheses and suggested instead that the ultimate cause of psychological disorders was traumatic experiences from early childhood, especially early sexual experiences. In some of his writings, he attributed those experiences to seductions by other children; in later reports, he said that the father or other adults had sexually abused the children. He had no evidence for any of his accusations; his patients did not even recall any of these events, but Freud inferred that they must have happened anyway. He put together parts of the patients’ dream reports, slips of the tongue, and so forth, and claimed that these all pointed to early sexual abuse. At

this point, Freud’s problems with evidence began to become serious. His ideas about inadequate sex or too much masturbation may sound ridiculous, but at least they were testable. But how was anyone to determine whether the events that Freud inferred from dreams and so forth had actually happened?

A few years later, he backed down from saying that psychological disorders resulted from childhood sexual abuse. According to Freud’s own description of these events, he realized that his patients had “misled” him into believing they had been sexually abused in early childhood; in fact, Freud said, their claims of sexual contact with their parents were “wishful phantasies” (Freud, 1925). Although he never fully developed his views of girls’ early sexual development, he was more explicit about boys: During early childhood, every little boy goes through an **Oedipus complex**; he *develops a sexual interest in his mother and competitive aggression toward his father*. (Oedipus—EHD-ah-puhs—a figure in an ancient Greek play by Sophocles, unknowingly murdered his father and married his mother.) In short, Freud said people become psychologically disturbed not because their parents sexually molested them, but because as children they had wanted to have sex with their parents and couldn’t.

Why did Freud switch from his idea about early sexual abuse to a theory about early sexual fantasies? According to one view (Masson, 1984), Freud was right the first time, and he simply lost the courage to defend his views about childhood sexual abuse. According to other scholars, however, Freud never had any justification for either set of views (Esterson, 1998; Powell & Boer, 1994; Schatzman, 1992). In his earliest writings, Freud had inferred his patients’ sexual abuse from their symptoms and dreams, despite their own denials of such sexual abuse. It was hardly fair, therefore, to complain that the patients had “misled” him into believing they had been sexually abused. When he switched to saying that the patients had had an Oedipus complex or other childhood sexual fantasies, he was again drawing inferences that his patients denied. (Freud considered his patients’ protests to be signs of emotional resistance and therefore confirmation that his interpretations were correct.)

Apparently, Freud’s main evidence for his interpretations was simply that he was able to construct a coherent story linking a patient’s symptoms, dreams, and so forth to the sexual fantasies that Freud inferred must have occurred (Esterson, 1993). In short, he did not distinguish between his “results” and his “interpretations.”

Further Examples of Casual Use of Evidence

Freud preferred complicated explanations to more parsimonious ones. At one point, his friend Wilhelm Fliess performed a surgical operation on a woman’s nose to try to relieve her stomach and menstrual pains—no, that’s not a misprint—and absentmindedly left half a meter of surgical

gauze in her nasal cavity after the operation. Her nose became infected and her face disfigured, and she almost bled to death. In a letter to Fliess, Freud suggested that the reason the woman continued to bleed for so long was that she had “an old wish to be loved in her illness” (Masson, 1985, Letter of May 4, 1896).

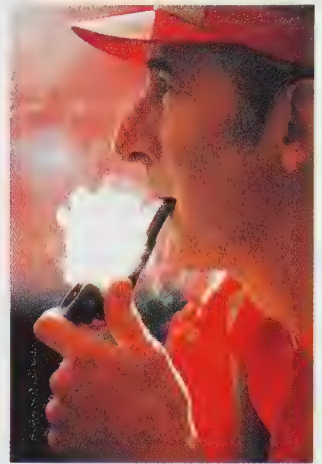
Another patient, known as Dora, came to Freud at her father’s insistence. Since about age 8, Dora had suffered from headaches, coughing, and shortness of breath, all of which Freud interpreted as psychological in nature. She had also avoided a certain friend of her father, Mr. “K,” ever since one episode when K grabbed Dora, then 14 years old, and kissed her on her lips. Freud wrote that Dora surely experienced sexual pleasure at this time, partly because K’s presumably erect penis rubbed up against her clitoris and excited it. Freud described Dora’s reaction of disgust as “. . . entirely and completely hysterical. I should without question consider a person hysterical in whom an occasion for sexual excitement elicited feelings that were preponderantly or exclusively unpleasurable” (Freud, 1905/1963, p. 44). He then proceeded to try to explain Dora’s hysterical behavior, tracing it to her love for both K and her father and her homosexual attraction to Mrs. K. As one reads Freud’s elaborate discussion, one wonders what was wrong with Dora’s simpler explanation—that she really was repulsed by the sexual advances of this middle-aged man.

Stages of Psychosexual Development in Freud’s Theory of Personality

Right or wrong, Freud’s theory is so well known that, as a student of psychology, you should know what these terms mean when you hear someone else talking about them. One of his central points was that psychosexual interest and pleasure begin in infancy. He used the term **psychosexual pleasure** in a broad sense to include *all strong, pleasant excitement arising from body stimulation*. He maintained that how we deal with our psychosexual development influences nearly all aspects of our personality.

Freud proposed that young children have sexual tendencies that resemble those of more primitive mammals. Just as nonhuman mammals respond sexually to sounds and smells that do not excite most adult humans, children respond sexually to stimulation of the mouth, the anus, and other body zones. Freud based his views on his own and his patients’ reconstructions of their childhood; he did not systematically observe children himself.

According to Freud (1905/1925), people have a *psychosexual energy*, which he called **libido** (lih-BEE-doh), from a Latin word meaning “desire.” Normally, libido is focused in an infant’s mouth and “flows” to other parts of the body as the child grows older. Children go through five stages of psychosexual development, each with a characteristic sexual focus that leaves its mark on the adult personality. If normal sexual development is blocked or frustrated at any stage, Freud said, part of the libido becomes **fixated** at



The beginning of psychosexual development is the oral stage, when infants enjoy stimulation of their mouth—which for them means sucking, swallowing, and biting. They like putting things in their mouths and gnawing on them. According to Freud, if normal sexual development is blocked at this stage, the child will grow up continuing to get much pleasure from drinking and eating, as well as from kissing and smoking. Perhaps this pipe smoker’s mother weaned him too quickly—or let him nurse too long. And perhaps such an explanation is incorrect. Like many of Freud’s ideas, this one is difficult to test.

that stage; that is, it *continues to be preoccupied with the pleasure area associated with that stage*. Table 13.3 summarizes these stages.

The Oral Stage In the **oral stage**, from birth through the first year or so (Freud was vague about the age limits of each stage), *the infant derives intense psychosexual pleasure from stimulation of the mouth, particularly while sucking at the mother’s breast*. In the later part of the oral stage, the infant begins to bite as well as suck. A person fixated at this stage continues to receive great pleasure from eating, drinking, and smoking and may also have lasting concerns with dependence and independence.

The Anal Stage At about 1 to 3 years of age, children enter the **anal stage**. At this time *they get psychosexual pleasure from stimulation of the anal sphincter, the muscle that controls bowel movements*. A person fixated at this stage goes through life “holding things back”—being orderly, stingy, and stubborn—or, less commonly, may go to the opposite extreme and become wasteful, messy, and destructive.

The Phallic Stage Beginning at about age 3, in the **phallic stage**, children begin to *play with their genitals*, and, according to Freud, become sexually attracted to the opposite-sex parent. Freud claimed that boys with a phallic fixation are afraid of being castrated; girls with such a fixation develop “penis envy.” These ideas have long been controversial; developmental psychologists almost never observe castration fear or penis envy in children.

TABLE 13.3 Freud's Stages of Psychosexual Development

STAGE (approximate ages)	SEXUAL INTERESTS AT THIS STAGE	EFFECTS OF FIXATION
<i>Oral Stage</i> (birth to 1 year)	Sucking, swallowing, biting	Lasting concerns with dependence and independence; pleasure from eating, drinking, and other oral activities
<i>Anal stage</i> (1 to 3 years)	Expelling feces, retaining feces	Orderliness, stinginess, stubbornness
<i>Phallic stage</i> (3 to 5 or 6 years)	Touching penis or clitoris; Oedipus complex	Difficulty feeling closeness. Males: fear of castration Females: penis envy
<i>Latency period</i> (5 or 6 to puberty)	Sexual interests suppressed	—
<i>Genital stage</i> (puberty onward)	Sexual contact with other people	—

The Latent Period From about age 5 or 6 until adolescence, Freud said, most children enter a **latent period** in which they *suppress their psychosexual interest*. At this time, they play mostly with peers of their own sex. The latent period is evidently a product of the culture and is not apparent in certain nonindustrialized societies.

The Genital Stage Beginning at puberty, young people *take a strong sexual interest in other people*. This is known as the **genital stage**. According to Freud, anyone who has fixated a great deal of libido in an earlier stage has little libido left for the genital stage. But people who have successfully negotiated the earlier stages can now derive primary satisfaction from sexual intercourse.

Evaluation of Freud's Stages Freud's theory makes such vague predictions that it is difficult to test (Grünbaum, 1986; Popper, 1986). When it has been tested, the results have been mostly unimpressive or inconclusive. For example, the characteristics of being orderly, stingy, and stubborn, which Freud described as due to anal fixation, do tend to correlate with one another, suggesting that they are part of a single personality type. However, we have no evidence that these attributes actually result from any aspect of toilet training (Fisher & Greenberg, 1977).

Freud's Description of the Structure of Personality

Personality, Freud claimed, consists of three aspects: the id, the ego, and the superego (Figure 13.11). (Actually, he used German words that mean *it*, *I*, and *over-I*. A translator used Latin equivalents instead of English words.) The **id** consists of *all our biological drives*, such as sex and hunger. The id demands immediate gratification. The **ego** is *the rational, decision-making aspect of the personality*. The **superego** contains *the memory of our parents' rules and prohibitions*,

such as, "Nice little boys and girls don't do that." Sometimes the id produces sexual or other motivations that the superego considers repugnant, thus evoking feelings of guilt. The ego may side with either the id or the superego; if it sides with the superego, it tries to avoid even thinking about the id's unacceptable impulses. Most psychologists today find it difficult to imagine the mind in terms of three warring factions and therefore regard Freud's description as only a metaphor.

CONCEPT CHECK

2. What kind of behavior would Freud expect of someone with a strong id and a weak superego? What behavior would you expect of someone with an unusually strong superego? (Check your answers on page 496.)

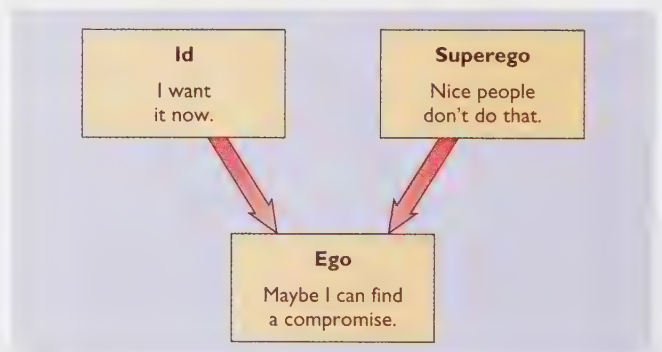


FIGURE 13.11 Freud described personality as a collection of three parts that are often in conflict with one another. The *id* asks for immediate gratification of biological urges, such as the sex drive. The *superego* counters these impulses with lists of rules that we learned from our parents. The *ego*, torn between these two forces, makes the decisions about what to do. According to Freud, we can understand certain people's behavior by assuming that their id is stronger than their superego or that their superego is stronger than their id.

Defense Mechanisms Against Anxiety

According to Freud, *the ego defends itself against conflicts and anxieties by relegating unpleasant thoughts and impulses to the unconscious*. Among the **defense mechanisms** that the ego employs are repression, denial, rationalization, displacement, regression, projection, reaction formation, and sublimation (Figure 13.12). Ordinarily, we are not aware of our own repressions, rationalizations, and so forth. Defense mechanisms are normal ways of suppressing anxiety and are often adaptive, becoming a problem only when they prevent a person from effectively dealing with reality.

Repression The defense mechanism of **repression** is *motivated forgetting*—the rejection of unacceptable thoughts, desires, and memories and their relegation to the unconscious. Repression is perhaps the most central concept in Freud's theory.

One example of repression would be a woman seeing someone beating another person to death and later not being able to remember what she saw. Another example: A man gives a speech, and several members of the audience raise serious objections to what he says; later, he forgets their objections.

Researchers have struggled to find unambiguous evidence to support the concept of repression. Investigators have exposed subjects to various unpleasant or threatening experiences, in the expectation that repression would interfere with their memories. However, whenever subjects did have trouble remembering the events, a variety of alternative explanations appear to be possible (Holmes, 1990). Outside the laboratory, repression is certainly not common in situations where we might expect it. As discussed in Chapter 7, kidnapping victims, children who watched the death of their own parents, and others who endured similar misery almost always remember the events intensely.

Denial The refusal to believe information that provokes anxiety ("This can't be happening") is called **denial**. Whereas repression is the motivated forgetting of certain information, denial is an assertion that the information is incorrect. For example, someone with a serious alcohol problem may insist, "I'm not an alcoholic. I can take it or leave it."

Rationalization When people attempt to prove that their actions are rational and justifiable and thus worthy of approval, they are using **rationalization**. For example, a

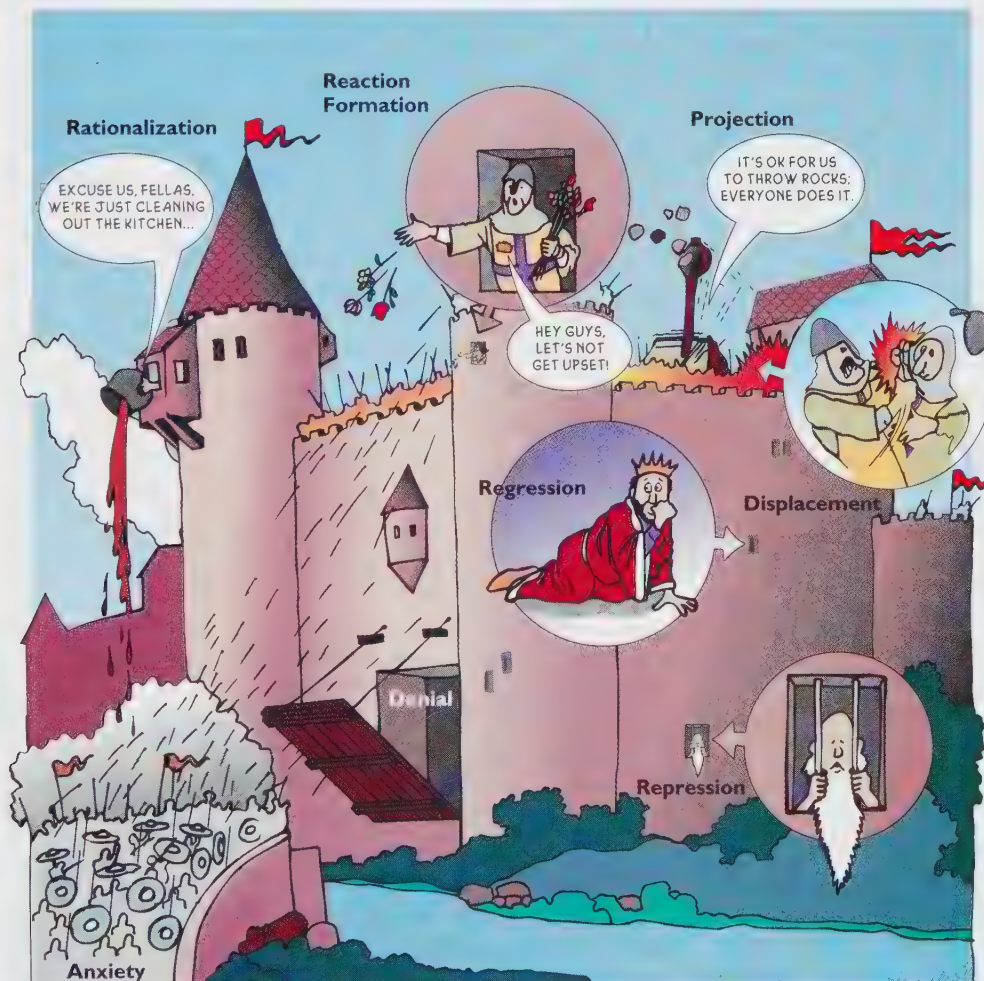


FIGURE 13.12 The ego—the rational I—has numerous ways of defending itself against anxiety, that apprehensive state named for the Latin word meaning “to strangle.” These defense mechanisms try to ignore or avoid facing unpleasant realities, and they are part of an internal battle that you fight against yourself.

student who wants to go to the movies instead of studying says, “More studying won’t do me any good anyway.” Someone who misses a deadline to apply for a job says, “I didn’t really want that job anyway.”

Displacement By diverting a behavior or thought away from its natural target toward a less threatening target, **displacement** lets people engage in the behavior they prefer without experiencing severe anxiety. For example, a man who is angry at his boss might come home and kick his dog. He really wants to kick his boss, but that would cause him too much anxiety. Or a student who is angry at her professor screams at her roommate.

Regression A return to a more juvenile level of functioning, **regression** is an effort to avoid the anxiety of facing one’s current role in life. By adopting a childish role, a person can escape responsibility and return to an earlier, perhaps more secure way of life. For example, after a new sibling is born, a 5-year-old child may start wetting the bed again. Following a divorce or a business setback, a man may resort to daydreaming, getting drunk, or other immature behaviors.

Projection The attribution of one’s own undesirable characteristics to other people is known as **projection**. When people project their own faults onto others, they do not necessarily deny that they possess those faults themselves (Holmes, 1978; Sherwood, 1981). However, by suggesting that the faults are widespread, they make them more acceptable and less anxiety provoking. For example, someone who secretly enjoys pornography might accuse other people of enjoying it.

Reaction Formation To reduce anxiety and to keep undesirable characteristics repressed, people may use **reaction formation** to present themselves as the opposite of what they really are. In Shakespeare’s play *Hamlet*, Gertrude says, “The lady protests too much, methinks.” People who insist too vehemently that something is “absolutely” true often harbor secret doubts about whether it is really true. For example, a man troubled by doubts about his religious faith may try to convert others to the faith. Someone with unacceptable aggressive tendencies may join a group dedicated to preventing violence against babies or animals. (Not everyone who proselytizes for a faith has deep doubts about it, of course, and not everyone who tries to prevent violence is secretly a violent person. Different people can have a variety of reasons for the same actions.)

Sublimation The transformation of sexual or aggressive energies into culturally acceptable, even admirable, behaviors is **sublimation**. According to Freud, sublimation enables a person to express the impulse without admitting its existence. For example, painting and sculpture may represent a sublimation of sexual impulses. Someone with unacceptable aggressive impulses may sublimate them by be-

coming a surgeon. However, if the true motives of a painter are sexual, as Freud proposed, they are well hidden indeed. Sublimation is the one proposed defense mechanism that is associated with socially constructive behavior.

CONCEPT CHECK

3. Match the Freudian defense mechanisms in the top list with the situations in the list that follows:

- | | |
|--------------------|-----------------------|
| 1. Repression | 5. Regression |
| 2. Denial | 6. Projection |
| 3. Rationalization | 7. Reaction formation |
| 4. Displacement | 8. Sublimation |

_____ a. A man who is angry with his neighbor goes hunting and kills a deer.

_____ b. Someone with a smoking habit insists there is no convincing evidence that smoking impairs health.

_____ c. A woman with doubts about her religious faith tries to convert others to her religion.

_____ d. A man who beats his wife writes a book arguing that people have an instinctive need for aggressive behavior.

_____ e. A woman forgets a doctor’s appointment for a test for cancer.

_____ f. Someone who has difficulty dealing with certain people resorts to pouting, crying, and throwing tantrums.

_____ g. A boss takes credit for a good idea suggested by an employee because, “It’s better for me to take the credit so that our department will look good and all the employees will benefit.”

_____ h. Someone with an unacceptable impulse to shout obscenities becomes a writer of novels.

(Check your answers on page 496.)

Manifestations of the Unconscious in Everyday Life

Freud believed that the unconscious made itself felt in nearly all aspects of ordinary life. Even an act that may be explained as “just a meaningless accident” reflects an unconscious motivation. For example, whenever one of Freud’s patients “forgot” an appointment, Freud assumed the patient did not want to keep it.

Freud also interpreted *slips of the tongue*, or what have come to be called “Freudian slips,” as revelations of unconscious thoughts or motives. If you said “I leave you” instead of “I love you,” Freud would assume that your error revealed an unconscious rejection of your professed love.

Today, psychologists believe that most slips of the tongue have multiple causes. Here is an example that is not really a slip of the tongue, but it helps to illustrate the point.

TRY IT YOURSELF

Ask someone to repeat the word “poke” ten times, about once every two seconds. At the end, quickly ask, “What do you call the white of an egg?” Your friend will almost certainly say “yolk,” instead of the correct answer, “albumin” (Baars, Fehling, LaPolla, & McGovern,

1997). Saying “poke” had primed your friend to say a similar-sounding word. Similarly, someone who reads “the Alpha and the Omega” as “the Alfalfa and the Omega” may have some interesting unconscious thoughts but has probably just confused two similar sounding words. Most slips of the tongue produce meaningless gibberish, and the occasional slips that seem meaningful could be just accidents.

Nevertheless, it is also possible that people’s motives do sometimes influence slips of the tongue. Let’s examine a clever attempt to test this idea.

WHAT IS THE EVIDENCE? *Freudian Slips*

How might one test the relation between people’s motives and their slips of the tongue? If we followed people around and recorded all their slips of the tongue, we would still have to guess or infer their hidden motives.

A second method would be to induce a motive—hunger, for example—for the purposes of an experiment. Then, we could ask the hungry subjects to read a certain passage, and we could count how many of their slips of the tongue had something to do with eating. We could repeat the experiment, inducing a different motive, to see whether people made corresponding slips.

However, most people make so few slips of the tongue that the experiment might go on for months with few results. We need a procedure that increases the frequency of slips of the tongue. Michael Motley and Bernard Baars (1979) devised just such a procedure.

Hypothesis When people are performing a difficult task on which they are likely to make slips of the tongue, the kinds of slips they make will depend on the motivations they are feeling at the moment.

Method The experimenters divided 90 male college students into three groups: a “sex” group, a “shock” group, and a control group. Those in the sex group were greeted by a very attractive female experimenter dressed in a sexy outfit and behaving seductively. Those in the shock group were met by a male experimenter who attached electrodes to

their arms and told them the electrodes were connected to a “random shock generator” that might give them one or more painful shocks at unpredictable times during the experiment. (No shocks were actually given.) Those in the control group were met by a male experimenter who attached no electrodes to them and made no mention of shocks. Thus, one group should have a heightened sexual motivation, one group should have a strong fear of shock, and one group should be concerned with neither sex nor shock.

The students watched a screen on which the experimenters flashed pairs of words or nonsense syllables, such as “HAT-RAM” and “RUF-GAM.” Each pair was flashed for 1 second. A buzzer sounded 0.4 second after each pair appeared, telling the students to speak aloud the *previous* pair. So, for example, after seeing “RUF-GAM,” they would have to say “HAT-RAM” (the previous pair), and remember “RUF-GAM” to say after the next pair. Students made slips of the tongue about 30% of the time. For example, they would say “HAT-RAM” as “RAT-HAM” and “RUF-GAM” as “GUF-RAM.”

Some of the syllable pairs were designed to promote sex-related slips. For example, “GOXI-FURL” and “LOOD-GECS” might be repeated as “FOXY-GIRL” and “GOOD-LEGS.” Other pairs were designed to promote shock-related slips. For example, “SHAD-BOCK” and “WOT-HIRE” might be pronounced as “BAD-SHOCK” and “HOT-WIRE.” Other pairs did not suggest any slips related to either sex or shock.

Results The men in the sex group made more than twice as many sex-related slips as shock-related slips. The opposite was true for the men in the shock group. Those in the control group made both types of errors about equally. Figure 13.13 illustrates the results.

Interpretation The results support Freud’s claim that a strong motivation can increase the frequency of slips of the tongue related to that motivation. Slips of the tongue may indeed sometimes reflect a person’s thoughts and desires.

But the results do not support Freud’s contention that hidden motivations are the main cause of slips of the tongue. Slips were common in this experiment because the task was so difficult. Having to say one pair while preparing

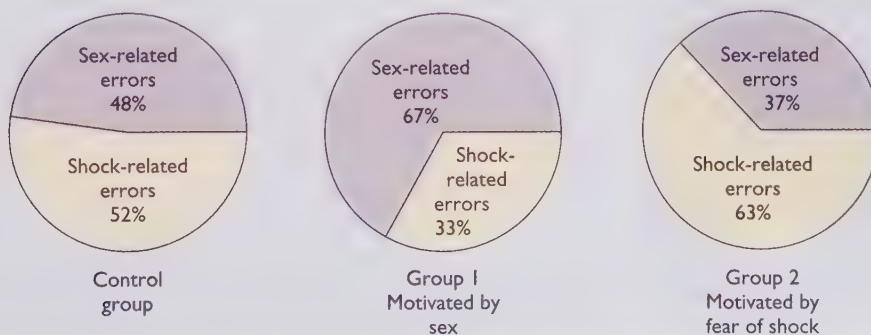


FIGURE 13.13 In the presence of an attractive woman, men made many sex-related slips of the tongue. When they were worried about shocks, they made shock-related slips of the tongue.

to say another produced conflict between the two pairs that a subject was trying to remember.



Freud's Legacy

Undeniably, Freud was a great pioneer in identifying new questions. The validity of his answers is quite controversial, however. He based his conclusions on inferences he drew from what his patients said and did, and he had no sure way of testing the validity of those inferences. A growing number of psychologists today contend that Freud imposed theories onto his data instead of drawing conclusions from the data.

It is possible, of course, for a visionary leader to induce correct theories based on weak evidence, and Freud could have been right even if his methods were faulty. How well his theories stand up today depends on who you ask, but the aspects of his theories that attract the fewest arguments are generalities such as the following:

- Our behavior is molded by earlier experiences dating back to childhood.
- Every behavior has multiple causes, and we are unaware of many of them.
- Someone can react strongly to a fairly normal experience because of its symbolic significance to that individual.
- We have a variety of ways of defending ourselves against anxiety.
- Talking about a psychological problem sometimes helps.

Compared to what Freud was trying to accomplish, however, giving him credit for such generalities is “damning by faint praise.” Freud thought he had devised a method to discover the contents of people’s unconscious minds. Such inferences might sometimes be correct; the problem, however, is to distinguish valid inferences from fanciful ones. On that point, many psychologists are most skeptical.



Freud's own couch is now part of our history. What is the future of psychoanalysts' couches? Psychoanalytic interpretations range from reasonable to doubtful; in any individual case, it is difficult or even impossible to judge the accuracy of an analyst's interpretation.



FIGURE 13.14 Karen Horney, a major neo-Freudian, revised some of Freud's theories and gave greater attention to cultural influences. She was a pioneer in the development of feminine psychology.

Neo-Freudians

Some psychologists, known as **neo-Freudians**, have *remained faithful to parts of Freud's theory while modifying other parts*. One of the most influential neo-Freudians was the German physician Karen Horney (HOR-nigh; 1885–1952; Figure 13.14), who believed that Freud had exaggerated the role of the sex drive in human behavior and had misunderstood the sexual motivations of women. She believed, for example, that the conflict between a child and his or her parents was a reaction to parental hostility and intimidation, not a manifestation of sexual desires. Horney contended that Freud had slighted the importance of cultural influences on personality and that he neglected to help his patients work out practical solutions to their problems. Still, Horney's views were more a revision than a rejection of Freud's theories. Other theorists, including Carl Jung and Alfred Adler, broke more sharply with Freud. Although some psychologists call Jung and Adler neo-Freudians, the followers of Jung and Adler do not. Each offered a very different, distinctly non-Freudian view of personality.

Carl Jung and the Collective Unconscious

Carl G. Jung (YOONG; 1875–1961; Figure 13.15), a Swiss physician, was an early member of Freud's inner circle. Freud regarded Jung as a son, the “heir apparent” or “crown prince” of the psychoanalytic movement, until their father-son relationship began to deteriorate (Alexander, 1982). At one point, Freud and Jung agreed to analyze each other's dreams. Freud described one of his dreams, but then refused to provide the personal associations that would enable Jung to interpret it, insisting that “I cannot risk my authority.”

Jung was more forthcoming. He described a dream in which he explored the upper stories of a house, then explored its basement, and finally discovering that the house had a subbasement, began to explore that. Jung thought the dream referred to his explorations of the mind. The top floor was the conscious mind; the basement was the unconscious; and the subbasement was a still deeper level of the unconscious, yet to be explored. Freud, however, insisted



FIGURE 13.15 Carl G. Jung rejected Freud's concept that dreams hide their meaning from the conscious mind: "To me dreams are a part of nature, which harbors no intention to deceive, but expresses something as best it can" (Jung, 1965).

that the dream referred to Jung's personal experiences and frustrations (Hannah, 1976).

Jung's own theory of personality incorporated some of Freud's ideas but put greater emphasis on people's search for a spiritual meaning in life and on the continuity of human experience, past and present. Jung believed that every person has not only a conscious mind and a "personal unconscious" (equivalent to Freud's "unconscious"), but also a collective unconscious. The personal unconscious represents a person's own experience. The **collective unconscious**, which is present at birth, represents *the cumulative experience of preceding generations*. Because all humans share a common ancestry, all have the same collective unconscious. The collective unconscious contains **archetypes**, which are *vague images that we all inherit from the experiences of our ancestors*. As evidence for this view, Jung pointed out that similar images emerge in the art of cultures throughout the world (Figure 13.16) and that similar themes emerge in various religions, myths, and folklore. Those images and themes also appear in dreams and in the hallucinations of people with severe psychological disorders.

Given what biologists now know of genetics, Jung's ideas are hard to defend. Having an experience does not change one's genes, and even if we did somehow develop genes that represented common human experiences, those genes would certainly vary among people, as other genes do. (Jung was quite insistent that the collective unconscious was the same for all people.) Jung's alternative to a genetic explanation was that perhaps archetypes exist on their own, independent of time, space, and brains. That is a difficult idea even to contemplate, much less test. In short, Jung's views of the collective unconscious and archetypes are vague and mystical, although a much modified version could be developed that would be scientifically testable (Neher, 1996).

Alfred Adler and Individual Psychology

Alfred Adler (1870–1937; Figure 13.17), an Austrian physician who, like Jung, had been one of Freud's early associates, broke with Freud because he believed Freud was overemphasizing the sex drive and neglecting other, more important influences on personality. They parted company in 1911, with Freud insisting that women experience "penis envy" and

with Adler contending that women were more likely to envy men's status and power. The two were never reconciled.

Adler founded a rival school of thought, which he called **individual psychology**. To Adler, this term did not mean "psychology of the individual." Rather, it meant "*invisible psychology*," a *psychology of the person as a whole rather than a psychology of parts*, such as id, ego, and superego. Adler emphasized the importance of conscious, goal-directed behavior and deemphasized (though he did not deny) unconscious influences.

Adler's Description of Personality

Several of Adler's early patients were acrobats who had had an arm or a leg damaged by a childhood illness or injury. These people were determined to overcome their disabilities, and they had worked hard to develop the strength and coordination they needed to perform as acrobats. Perhaps, Adler surmised, people in general try to overcome their weaknesses and to transform them into strengths (Adler, 1932/1964).

As infants, Adler pointed out, we are small, dependent creatures who strive to overcome our inferiority. People who do not succeed go through life with an **inferiority complex**, an *exaggerated feeling of weakness, inadequacy, and helplessness*. Even those who do manage to overcome their feelings of inferiority persist in their efforts to achieve.

According to Adler, everyone has a natural **striving for superiority**, a *desire to seek personal excellence and fulfillment*. Each person creates a **style of life**, or *master plan for achieving a sense of superiority*. That style of life may be directed toward success in business, sports, politics, or another competitive activity. Or it may be directed toward "success" of a different sort: For example, someone who withdraws from life may gain a sense of accomplishment or superiority from being uncommonly self-sacrificing. Someone who constantly complains about real or imagined illnesses or disabilities may, by demanding help from friends and family, win a measure of control or superiority over them. Or someone may commit crimes to savor the attention that they bring.

Adler recognized that people are not always aware of their own style of life and the assumptions behind it and may fail to realize that the real motive behind a word or action is to manipulate others. They may engage in self-defeating behavior because they have not admitted to themselves what their goals really are. Adler tried to determine people's real motives. For example, he would ask someone who complained of a backache, "How would your life be different if you could get rid of your backache?" Those who said they would become more active were presumably suffering from real ailments that they were trying to overcome. Those who said they could not imagine how their life would change, or said only that they would get less sympathy from others, were presumably suffering from psychologically caused ailments or, at least, were exaggerating their discomfort.

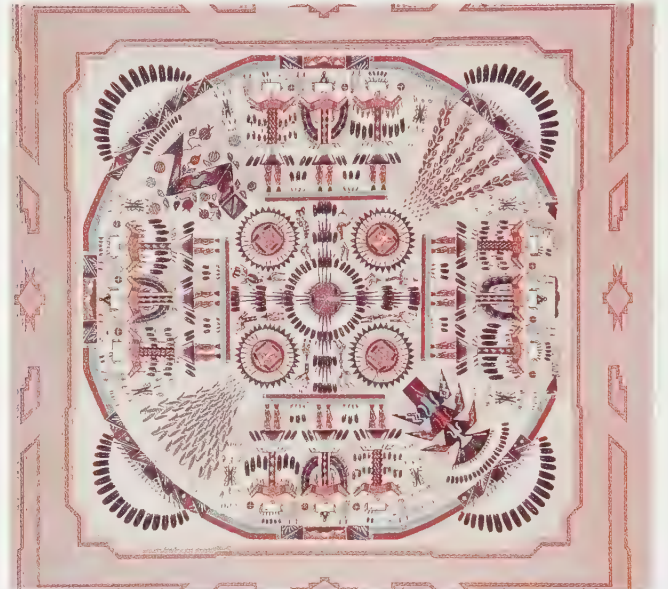


FIGURE 13.16 Carl Jung was fascinated with the similar images that show up in the artwork of different cultures. One recurring image is the mandala, which Jung believed to be a symbol of the self's striving for unity and wholeness. These mandalas are (upper left) a Hindu painting from Bhutan; (upper right) a mosaic from Beth Alpha Synagogue, Israel; (lower left) a tie-dye tapestry created in the United States; and (lower right) a Navajo sand painting from the southwestern United States.

CONCEPT CHECK

4. In Adler's theory, what is the relationship between striving for superiority and style of life? (Check your answer on page ...)

Adler's View of Psychological Disorders

Any personality based on a selfish style of life is unhealthy, Adler (1928/1964) said. People's need for one another requires that they develop a **social interest**, a *sense of solidarity and identification with other people*. People

with a strong social interest strive for superiority in a way that contributes to the welfare of the whole human race, not just to their own welfare. They want to cooperate with other people, not to compete. In equating mental health with a strong social interest, Adler saw mental health as a positive state, not just the absence of impairments.

In Adler's view, people with psychological disorders are not suffering from an "illness." Rather, they have set immature goals, are following a faulty style of life, and show little social interest. Their response to new opportunity is "Yes, but . . ." (Adler, 1932/1964). They are striving for superiority in ways that are useless to themselves and to others.



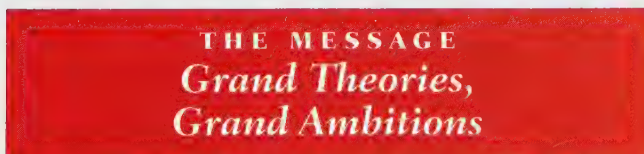
FIGURE 13.17 Like Horney, Alfred Adler thought that Freud overemphasized the sex drive. Adler was very interested in feelings of self-esteem. He was the first to talk about the possibility of a “healthy” personality as not just a personality free from disorders. According to Adler, the key to a healthy personality was “social interest,” a desire for the welfare of other people.

For example, one of Adler's patients was a man who lived in conflict with his wife because he was constantly trying to impress her and dominate her (Adler, 1927). When discussing his problems, the man revealed that he had been very slow to mature physically and had not reached puberty until he was 17 years old. Other teenagers had ignored him and had treated him like a child. He was now a physically normal adult, but he was overcompensating for those years of feeling inferior by trying to seem bigger and more important than he really was.

Adler tried to get patients to understand their own style of life and to correct the faulty assumptions on which they had based their lives. He urged them to strengthen their social interest and to strive for superiority in ways that would benefit both themselves and others.

Adler's Legacy

Adler's influence on psychology exceeds his fame. His concept of the “inferiority complex” has become part of the common culture. He was the first to talk about mental health as a positive state rather than as merely the absence of impairments. Many later forms of therapy drew upon Adler's innovations, especially his emphasis on the assumptions underlying a patient's behavior. Humanistic psychologists followed Adler by urging people to take responsibility for their own behavior and for modifying their style of life.



Freud, Jung, and Adler lived and worked in an earlier era that was apparently more conducive to building major theories of how personality fits together. It is easy to fault these theorists for jumping to conclusions and for their other shortcomings. Still, one should admire the scope of what they were trying to accomplish and the influence that they had on later investigators. Previously, personality was entirely in the realm of philosophers and novelists. The grand theory builders made personality a topic for research and progress.

S U M M A R Y

- ★ **Freud.** Sigmund Freud, the founder of psychoanalysis, proposed that human behavior is greatly influenced by unconscious thoughts and motives and that much of what we do and say has hidden meanings. (page 485)
- ★ **Freud's psychosexual stages.** Freud believed that many unconscious thoughts and motives are sexual in nature. He proposed that people progress through stages or periods of psychosexual development—oral, anal, phallic, latent, and genital—and that frustration at any one stage can lead to a lasting fixation of the libido at that stage. (page 487)
- ★ **Slips of the tongue.** Unconscious thoughts influence many aspects of everyday life, including slips of the tongue, although other influences may be more important. (page 490)
- ★ **Jung.** Carl Jung believed that all people share a “collective unconscious” that represents the entire experience of humanity. (page 492)
- ★ **Adler.** Alfred Adler proposed that people's primary motivation is a striving for superiority. Each person adopts his or her own “style of life,” or method of striving. (page 493)
- ★ **Adler's view of a healthy personality.** According to Adler, the healthiest style of life is one that emphasizes “social interest”—that is, concern for the welfare of others. (page 494)

Suggestions for Further Reading

- Adler, A. (1954). *Understanding human nature*. Greenwich, CT: Fawcett. (Original work published 1927.) Adler's most general and most popular book.
- Freud, S. (1924). *Introductory lectures on psychoanalysis*. New York: Boni & Liveright. Available in various paperback editions. This is Freud's attempt to describe the fundamentals of his theory to a general audience.
- Macmillan, M. (1997). *Freud evaluated*. Cambridge, MA: MIT Press. A thorough criticism of Freud's theories and his use of evidence.

Terms

- catharsis** the release of pent-up emotional tension (page 485)
- psychoanalysis** Freud's approach to personality, based on the interplay of conscious and unconscious forces (page 485)
- unconscious** according to Freud, the repository of memories, emotions, and thoughts—often illogical thoughts—that affect our behavior even though we cannot talk about them (page 485)
- Oedipus complex** according to Freud, a young boy's sexual interest in his mother accompanied by competitive aggression toward his father (page 486)
- psychosexual pleasure** according to Freud, any strong, pleasant enjoyment arising from body stimulation (page 487)
- libido** in Freud's theory, psychosexual energy (page 487)
- fixation** in Freud's theory, a persisting preoccupation with an immature psychosexual interest as a result of frustration at that stage of psychosexual development (page 487)
- oral stage** Freud's first stage of psychosexual development; here, psychosexual pleasure is focused on the mouth (page 487)

anal stage Freud's second stage of psychosexual development; here, psychosexual pleasure is focused on the anus (page 487)

phallic stage Freud's third stage of psychosexual development; here, psychosexual interest is focused on the penis or clitoris (page 487)

latent period according to Freud, a period in which psychosexual interest is suppressed or dormant (page 488)

genital stage Freud's final stage of psychosexual development, in which sexual pleasure is focused on sexual intimacy with others (page 488)

id according to Freud, the aspect of personality that consists of all our biological drives and demands for immediate gratification (page 488)

ego according to Freud, the rational, decision-making aspect of personality (page 488)

superego according to Freud, the aspect of personality that consists of memories of rules put forth by one's parents (page 488)

defense mechanism a method employed by the ego to protect itself against anxiety caused by the conflict between the id's demands and the superego's constraints (page 489)

repression motivated forgetting; the relegation of unacceptable impulses or memories to the unconscious (page 489)

denial the refusal to believe information that provokes anxiety (page 489)

rationalization attempting to prove that one's actions are rational and justifiable and thus worthy of approval (page 489)

displacement the diversion of a thought or behavior away from its natural target toward a less threatening target (page 490)

regression the return to a more juvenile level of functioning as a means of reducing anxiety or in response to emotionally trying circumstances (page 490)

projection the attribution of one's own undesirable characteristics to other people (page 490)

reaction formation presenting oneself as the opposite of what one really is, in an effort to reduce anxiety (page 490)

sublimation the transformation of sexual or aggressive energies into culturally acceptable, even admirable, behaviors (page 490)

neo-Freudians personality theorists who have remained faithful to parts of Freud's theory while modifying other parts (page 492)

collective unconscious according to Jung, an inborn level of the unconscious that symbolizes the collective experience of the human species (page 493)

archetypes according to Jung, vague images inherited from our ancestors and contained in the collective unconscious (page 493)

individual psychology the psychology of the person as an indivisible whole, as formulated by Adler (page 493)

inferiority complex an exaggerated feeling of weakness, inadequacy, and helplessness (page 493)

striving for superiority according to Adler, a universal desire to seek personal excellence and fulfillment (page 493)

style of life according to Adler, a person's master plan for achieving a sense of superiority (page 493)

social interest a sense of solidarity and identification with other people (page 494)

Answers to Concept Checks

- Someone with a strong id and a weak superego could be expected to give in to a variety of sexual and other impulses that other people would inhibit. Someone with an unusually strong superego would be unusually inhibited and dominated by feelings of guilt. (page 488)
- a. 4, displacement; b. 2, denial; c. 7, reaction formation; d. 6, projection; e. 1, repression; f. 5, regression; g. 3, rationalization; h. 8, sublimation. (page 490)
- In Adler's theory, a person's style of life is a method of striving for superiority. (page 494)

Web Resources

The Freud Web

www.stg.brown.edu/projects/hypertext/landow/HTatBrown/freud/Freud_OV.html

David B. Stevenson offers an in-depth look at Freud the man and at major elements of Freudian theory.

The Jung Index

www.jungindex.net/

One of the best psychology sites on the Web includes a detailed introduction to Jung's theory, a glossary of Jungian terms, a newsletter, and other valuable features.

Alfred Adler Institute of San Francisco

ourworld.compuserve.com/homepages/hstein/

The definitive Adler site includes dozens of articles about classical Adlerian psychology, how to become an Adlerian therapist, biographical sketches of Adler and prominent Adlerian therapists, and more.

Personality Traits

Is personality consistent over time and from one situation to another?

What are personality traits?

In many ways, all rocks are the same. If you want to predict when a dropped rock will hit the ground, you do not need to know what kind of rock it is. If you skip a rock across a lake, throw it against a window, or use it to crack open a coconut, you can predict what will happen if you know the size and shape of the rock. You don't need to know much more.

For other purposes, however, you can't treat all rocks the same way. If you want to predict what will happen if you

run an electric current through it, you have to know how much metal is in the rock. If you want to determine a fair sale price for the rock, you need to know whether it is a diamond or a piece of granite.

Similarly, people resemble one another in some ways and differ in others. Psychologists investigate individual differences in two ways, called the nomothetic and the idiographic approaches. The word *nomothetic* (NAHM-uh-THEHT-ick) comes from the Greek *nomothetes*, meaning "legislator"; the **nomothetic approach** seeks general laws about how an aspect of personality affects behavior, often based on statistical comparisons of large groups of people. For example, we might make the nomothetic statement that people with a more *Machiavellian* personality tend to be highly manipulative and that people with a less Machiavellian personality tend to cooperate with others. We could apply that statement fairly directly to anyone, simply by measuring that person's degree of Machiavellianism.

In contrast, the word *idiographic* is based on the root *idio-*, meaning "individual." The **idiographic approach** concentrates on *intensive studies of individuals* (Allport, 1961). For example, certain psychologists have studied how people's life goals affect their moods and their reactions to various events. Because people have different goals, the investigators draw carefully qualified conclusions about how people with one kind of goal behave and how people with another kind of goal behave (Emmons, 1991). The idiographic approach can lead to a conclusion that applies to more than just one person, but it is not meant to generalize to the whole population.

Personality Traits and States

Meteorologists distinguish between climate (the usual conditions) and weather (the current conditions). For example, the climate in London, England, is moister and cooler than the climate in south Texas, but on a given day the weather could be warm in London and cool in Texas. Similarly, psychologists distinguish between long-lasting personality conditions and temporary fluctuations.

A *consistent, long-lasting tendency in behavior, such as shyness, hostility, or talkativeness*, is known as a **trait**. In contrast, a **state** is a *temporary activation of a particular behavior*. People's behavior varies from time to time because they are in different states. For example, stage fright is a state; general nervousness is a trait. Being quiet in a library is a state; being quiet most of the time is a trait. A trait is like a climate condition that manifests itself in the long run, though not at every moment.

Note that both traits and states are descriptions of behavior, not explanations. To say that someone is nervous or quiet does not explain the person's behavior; it merely tells us what we are trying to explain.



Actors, such as this man playing the role of a woman in Japanese Kabuki theater, can present personalities that are very different from their private personalities. All of us occasionally display temporary personality states different from our usual selves.

CONCEPT CHECK

5. Suppose someone becomes nervous as soon as he sits down in a dentist's chair. Is this experience "trait anxiety" or "state anxiety"? (Check your answer on page 502.)

The Search for Broad Personality Traits

If we choose to describe people's personalities in terms of traits, how many personality traits should we identify? One way to begin is by examining our language. Presumably, if a particular personality trait is really important, any human language should have a word for it. Although this assumption could be wrong, it seems reasonable, considering how much attention people pay to other people's personalities.

Gordon Allport and H. S. Odbert (1936) plodded through an English dictionary and found almost 18,000 words that might be used to describe personality. Using this list, psychologists deleted words that were merely evaluations (such as *pleasant* or *nasty*) and terms referring to temporary states (such as *confused*). In the remaining list, they looked for clusters of synonyms, such as *affectionate*, *warm*, and *loving*, and kept only one of these terms. When they found opposites, such as *honest* and *dishonest*, they also kept just one of the terms. (*Honesty* and *dishonesty* are different degrees of a single trait, not two separate traits.) After eliminating synonyms and antonyms, Raymond Cattell (1965) narrowed the original list down to 35 traits.

The "Big Five" Personality Traits

Although some of the 35 personality traits that Cattell identified are not exactly synonyms or antonyms of one another, many of them overlapped enough to suggest that they were not independent traits. Furthermore, for practical purposes, psychologists usually prefer to deal with a smaller number. Remember the principle of parsimony from Chapter 2: If we can explain most aspects of personality with, say, 5 or 10 traits, we do not need to measure 35 or more.

To determine which traits correlate with one another, psychologists use a method called *factor analysis*. For example, if measurements of *warmth*, *gregariousness*, and *assertiveness* correlate strongly with one another, we can group them together into a single trait. But if this combined trait does not correlate highly with *self-discipline*, then we should distinguish *self-discipline* as a separate trait.

Using this approach, many researchers have found that they can describe most of the variance in human personality with what they call the **big five personality traits**: *neuroticism*, *extraversion*, *agreeableness*, *conscientiousness*, and *openness to new experience* (McCrae & Costa, 1987). These five factors offer a powerful description of personality



The Japanese artist Morimura Yasumasa reconstructs famous paintings, substituting his own face. Some people love his work; others dislike it or object to the whole idea. People high in "openness to experience" delight in new, unusual forms of art, literature, and music.

because each pertains to behavior in a wide variety of situations and because each of them correlates only weakly with any of the others. The big five dimensions are described below (Costa, McCrae, & Dye, 1991). Note that the first two, neuroticism and extraversion, are the "biggest" of the big five. Even psychologists who are skeptical of the big five model agree that neuroticism and extraversion are powerful traits that influence much of human behavior (Block, 1995).

Neuroticism is a tendency to experience unpleasant emotions relatively easily. People who test high in neuroticism are relatively likely to experience anxiety, hostility, depression, self-consciousness, and impulsiveness. The opposite of neuroticism is emotional stability or self-control.

Extraversion is a tendency to seek stimulation and to enjoy the company of other people. The opposite of extraversion is introversion. Extraversion is associated with warmth, gregariousness, assertiveness, impulsiveness, and a need for excitement. People high in extraversion enjoy meeting new people and tend to be happy under most circumstances (Francis, Brown, Lester, & Philipchalk, 1998). However, they are also more prone than other people to heavy use of alcohol, on the average (Martsh & Miller, 1997), and to risk taking in general.

Agreeableness is a *tendency to be compassionate toward others and not antagonistic*. It implies a concern for the welfare of other people, closely related to Adler's concept of social interest. People high in agreeableness generally trust other people and expect other people to trust them.

Conscientiousness is a *tendency to show self-discipline, to be dutiful, and to strive for achievement and competence*. Research has shown that people high in conscientiousness tend to be good workers on almost any job (Barrick & Mount, 1991). They are likely to complete whatever task they say they will perform.

Openness to experience, an uncommon trait in many populations, is a *tendency to enjoy new intellectual experiences and new ideas*. Someone high in this trait would be likely to enjoy modern art, unusual music, thought-provoking films and plays and would enjoy meeting different kinds of people and exploring new ideas and opinions (McCrae, 1996).

CONCEPT CHECK

- Some psychologists suggest that we should divide extraversion into two traits—which they call *ambition* and *sociability*—changing the big five into the big six. How should psychologists determine whether to do so? (Check your answer on page 502.)

Pros and Cons of the Big Five Description

To demonstrate that the “big five” traits provide a useful description for human personality, researchers must demonstrate that these five traits are nearly independent of one another and that they account for a major portion of the personality variation in many population samples, including both sexes, a wide range of ages, and a variety of cultures. For the most part, researchers have tested this hypothesis by repeating their original procedure on new samples of people. That is, they administer a questionnaire containing many personality descriptions (e.g., *friendly*, *competitive*, *shy*, and so forth) and ask people to rate themselves or other people on each description. Then, the psychologists conduct a factor analysis to determine which descriptions correlate with one another.

Support for the Big Five Description Many studies using this approach have concluded that the five-factor description is satisfactory for characterizing personality in both men and women, and that (in translation) the description also works in a variety of languages and cultures (McCrae & Costa, 1997). Certain differences do occur among cultures, however. For example, the research sometimes finds that one or another of the personality traits, such as openness, does not show enough individual variation to be

useful in some cultures (John, 1990). Also, some studies have found five personality traits to be useful—but five somewhat *different* traits from the more-established big five (Ormerod, McKenzie, & Woods, 1995). Psychologists need to make a judgment about whether these discrepancies create a serious problem, but most have concluded that the five-factor model works “well enough” across cultures.

How consistent does personality remain with age, as described by the five traits? As a rule, personalities seldom change drastically during adulthood (Costa & McCrae, 1994). We do find a few trends in personality as people grow older. Generally, between ages 20 and 30, most people decline in the “thrill-seeking” aspect of extraversion (McCrae & Costa, 1994). Over the entire course of adulthood, most people show a slight increase in agreeableness and conscientiousness and a moderate decrease in neuroticism, extraversion, and openness. The decrease in openness relates to the common observation that young adults are much more likely than older adults to enjoy new types of music and to try new kinds of foods, new styles of clothing, and so forth (Sapolsky, 1998). (Who listens more to new popular music—you or your parents? Who listens more to the “oldies” radio station?)

In some cases, people make a deliberate effort to change some aspect of their personality—to overcome an addiction or to decrease their neuroticism. Although changes can and do occur, they are almost always slow and difficult (DiClemente, 1994). In short, the personality of a young adult is likely to mature with age, but it is not likely to change drastically.

Criticisms of the Big Five Description In spite of the apparent success of the big five personality traits, several prominent psychologists caution against its wholehearted acceptance.

First problem: Nearly all of the data so far depend on written responses to questionnaires. Maybe what people say on questionnaires captures all the important dimensions of their personalities, but maybe it doesn't.

Second problem: The five-factor structure is theoretically unsatisfying. We have neither a psychological nor a biological explanation for why humans should have five main personality traits instead of another number, or why, if there are five, they should be these particular five (Block, 1995; Eysenck, 1992; Zuckerman, 1992).

Third problem: Choosing precisely five traits is arbitrary. Remember that Raymond Cattell initially identified 35 traits; factor analysis suggests that we can combine those 35 into 5 because various groups of the 35 traits correlate highly with one another. Still, the groups do not correlate perfectly. Some psychologists believe that we could describe personality more adequately with 7 or 8 traits (Block, 1995; Cloninger, 1994), or even with Cattell's original 35. Still other psychologists, noting a positive correlation between extraversion and openness and a negative correlation between neuroticism and conscientiousness, believe we could get by with 3 factors instead of 5 (Eysenck, 1992).

Overall, how should we evaluate the five-factor description? The answer depends on our purposes. If we are interested in a good, practical way to describe a great deal of the variation in human personality while using only a few terms, the five-factor description works well. However, if we are interested in a theoretical understanding of personality, it is an overstatement to call the five-factor description a fact of nature (as some have done).

The Origins of Personality

A description of personality differences is not an explanation. The question is: What makes some people more extraverted, neurotic, agreeable, conscientious, or open than other people are?

The two major categories of influence are heredity and environment. To measure these influences, researchers have relied mostly on two kinds of data. First, they compare the similarities between monozygotic (identical) twins and dizygotic (fraternal) twins. As Figure 13.18 shows, five studies conducted in separate locations indicated much greater similarities in extraversion between monozygotic pairs than between dizygotic pairs (Loehlin, 1992). However, such results are not easy to interpret. Some dizygotic twins make

a special effort to highlight their distinctiveness, and their parents also encourage them to be different from each other. Therefore, twin studies probably overstate the role of heredity in personality (Saudino, 1997).

Second, researchers compare the personalities of parents, their biological children, and their adopted children. As Figure 13.19 shows, parents' levels of extraversion correlate moderately with those of their biological children and hardly at all with those of their adopted children. Similarly, biologically related brothers or sisters growing up together resemble each other moderately; unrelated children adopted into the same family do not develop similar personalities (Loehlin, 1992). The results shown in Figures 13.18 and 13.19 pertain to extraversion; similar studies provide a largely similar pattern for neuroticism and other personality traits (Heath, Neale, Kessler, Eaves, & Kendler, 1992; Loehlin, 1992; Viken, Rose, Kaprio, & Koskenvuo, 1994).

Overall, these results indicate a moderate influence of hereditary factors, although no one knows exactly how they contribute. One possibility is that genetically determined differences in neurotransmitter receptors influence how strongly people respond to new stimuli, thereby predisposing some people to be more sensation-seeking or extraverted than others (Benjamin et al., 1996; Blum, Cull, Braverman, & Comings, 1996; Ebstein et al., 1996). However, we need

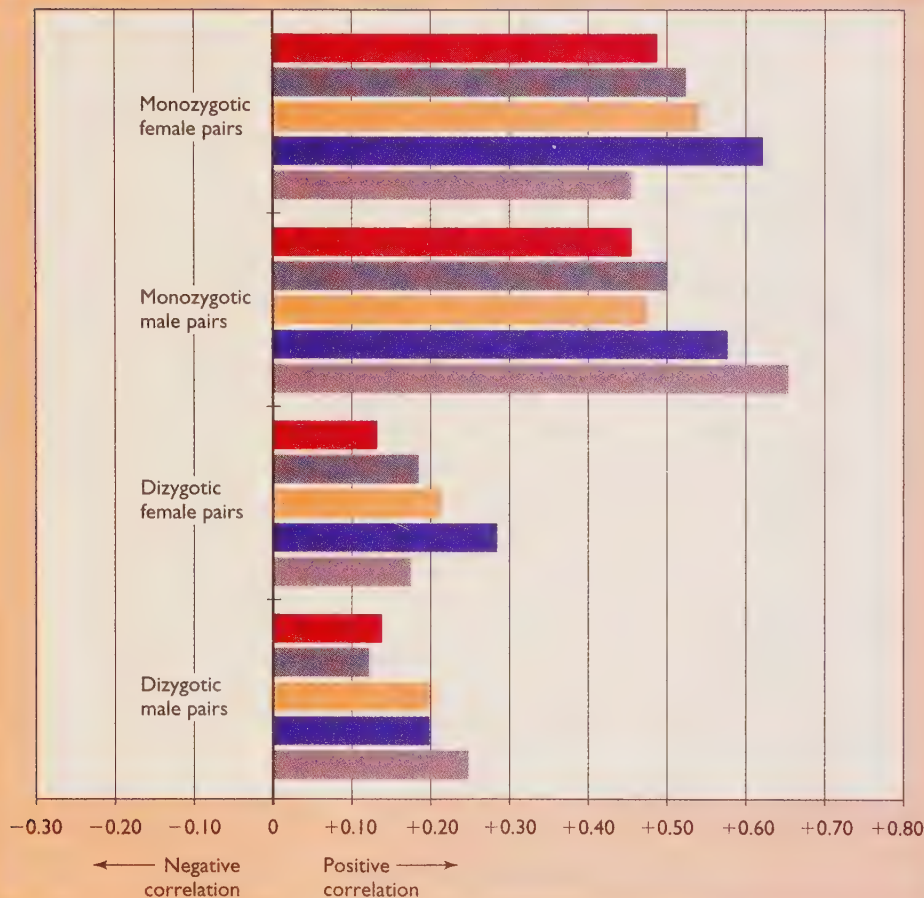
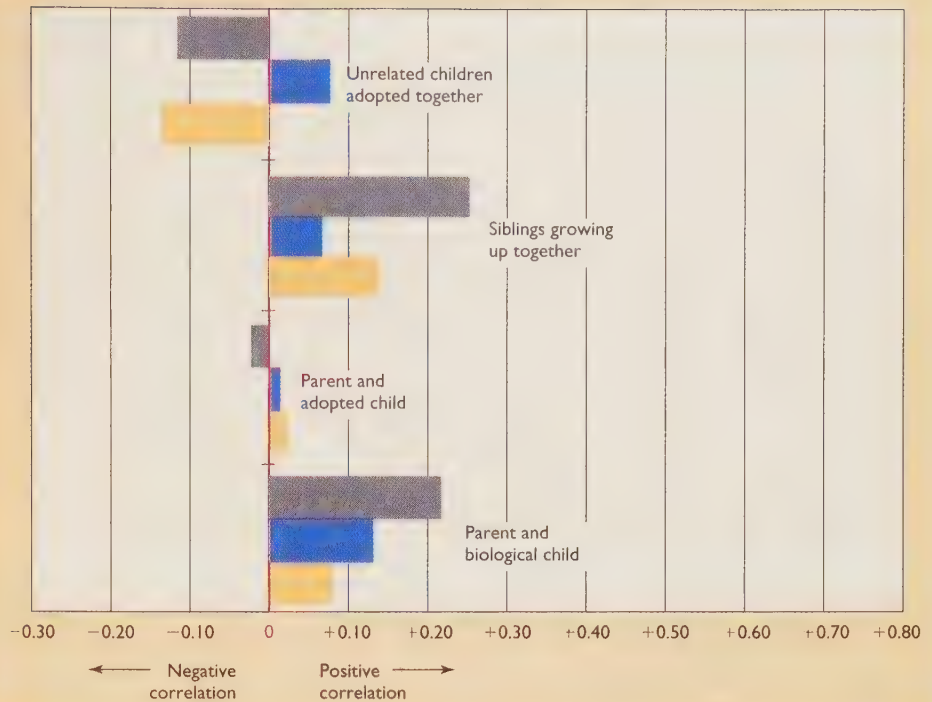


FIGURE 13.18 Five studies—conducted in Great Britain, the United States, Sweden, Australia, and Finland—found larger correlations between the extraversion levels of monozygotic (MZ) twins than those of dizygotic (DZ) twins. (Based on data summarized by Loehlin, 1992.)

FIGURE 13.19 Three studies—from Britain, Minnesota, and Texas—measured extraversion in members of hundreds of families. Each found moderate positive correlations between parents and their biological children and between pairs of biologically related brothers and sisters. However, all found low or even negative correlations between parents and adopted children or between adopted children living in the same family. (Based on data summarized by Loehlin, 1992.)



far more research to explore this and other possible links between genes and behavior.

The nongenetic contributors are even less understood. Even children who grow up with their biological parents resemble their parents in personality only to a moderate degree (Bratko & Marušić, 1997), and the personality correlations between adopted children and adoptive parents are even lower, sometimes even zero or negative. Evidently, children learn rather little of their personality by imitating their parents. Many researchers believe that much of the variation among people's personalities relates to the **unshared environment**, *the aspects of environment that differ from one individual to another, even within a family*. Unshared environment is, because of its idiosyncratic nature, very difficult to investigate. To some extent, unshared environment actually reflects genetic influences: People with various predispositions choose different activities and thereby create individual environments for themselves (Saudino, 1997).

THE MESSAGE The Challenges of Classifying Personality

Just as rocks are the same in some ways and different in others, personalities are the same in some regards and different in others. We find it useful to categorize personalities along a few dimensions, just as we find it useful to talk about different types of rocks.

In some ways, however, the analogy between classifying rocks and classifying personalities breaks down. The

problem is not just that every individual personality is different from every other; after all, every piece of granite is a little different from every other piece, too. Rather, the problem is that human personalities change from one situation to another far more than the properties of rocks do. As well as you know your closest friends and relatives, aren't you still sometimes surprised by how one of them acts in a new situation? Aren't you sometimes surprised even at your own behavior? The complexities of human personality make this area of research particularly challenging.

S U M M A R Y

- ✳ **Nomothetic and idiographic laws.** Psychologists seek both nomothetic laws, which apply to all people, and idiographic laws, which apply to individual differences. (page 497)
- ✳ **Traits and states.** Traits are personality characteristics that persist over time; states are temporary changes in behavior in response to particular situations. (page 497)
- ✳ **Five major traits.** Psychologists seek a short list of traits that describes as much of behavior as possible. Much can be explained by these five traits: neuroticism, extraversion, openness to new experience, agreeableness, and conscientiousness. (page 498)
- ✳ **Determinants of personality.** Studies of twins and adopted children indicate that heredity contributes to part of the observed differences in personality. Family environment evidently contributes rather little. Much of the variation in personality may be due to unshared environment, the special experiences that vary from one person to another even within a family. (page 500)

Suggestion for Further Reading

Pervin, L. A. (1990). *Handbook of personality*. New York: Guilford Press. Contains summaries of theory and research on a wide variety of topics related to personality.

Terms

- nomothetic approach** an approach to the study of individual differences that seeks general laws about how an aspect of personality affects behavior (page 497)
- idiographic approach** an approach to the study of personality differences that concentrates on intensive studies of individuals (page 497)
- trait** a consistent, long-lasting tendency in behavior (page 497)
- state** a temporary activation of a particular behavior (page 497)
- big five personality traits** five traits that account for a great deal of human personality differences: neuroticism, extraversion, agreeableness, conscientiousness, and openness to new experience (page 498)
- neuroticism** the tendency to experience unpleasant emotions relatively easily (page 498)
- extraversion** the tendency to seek stimulation and to enjoy the company of other people (page 498)
- agreeableness** the tendency to be compassionate toward others and not antagonistic (page 499)
- conscientiousness** the tendency to show self-discipline, to be dutiful, and to strive for achievement and competence (page 499)

- openness to experience** the tendency to enjoy new intellectual experiences, the arts, fantasies, and anything that exposes a person to new ideas (page 499)
- unshared environment** the aspects of environment that differ from one individual to another, even within a family (page 501)

Answers to Concept Checks

5. This nervousness is state anxiety, because it is evoked by a particular situation. Trait anxiety is a strong tendency to become nervous in many situations. (page 498)
6. They should determine whether measures of ambition correlate strongly with measures of sociability. If they do, then ambition and sociability can be considered two aspects of a single trait, extraversion. If they do not correlate strongly, then they are indeed separate personality traits. (page 499)

Web Resources

Personality Traits And Heart Disease Risk

www.pslgroup.com/dg/358ae.htm

This is a news report of a study that found that personality traits can increase or decrease the risk of heart attack. The trait that increases the risk, and who it affects most, may surprise you.

Personality Measures and the Big 5

fas.psych.nwu.edu/perproj/theory/big5.table.html

The different terms used by 16 different personality theorists to describe dimensions of personality are organized into five broad dimensions in this tabular comparison.

Personality Assessment

How can we measure personality?

How can we use measurements of personality?

A new P. T. Barnum Psychology Clinic has just opened at your local shopping mall and is offering a Grand Opening Special on personality tests. You have always wanted to know more about yourself, so you sign up. Here is Barnum's true-false test:

Questionnaire for Universal Assessment of Zealous Youth (QUAZY)

- | | |
|--|-----|
| 1. I have never met a cannibal I didn't like. | T F |
| 2. Robbery is the only major felony I have ever committed. | T F |
| 3. I eat "funny mushrooms" less frequently than I used to. | T F |
| 4. I don't care what people say about my nose-picking habit. | T F |
| 5. Sex with vegetables no longer disgusts me. | T F |
| 6. This time I am quitting glue-sniffing for good. | T F |
| 7. I generally lie on questions like this one. | T F |
| 8. I spent much of my childhood sucking on telephone cords. | T F |
| 9. I find it impossible to sleep if I think my bed might be clean. | T F |
| 10. Naked bus drivers make me nervous. | T F |
| 11. Some of my friends don't know what a rotten person I am. | T F |
| 12. I usually find laxatives unsatisfying. | T F |
| 13. I spend my spare time playing strip solitaire. | T F |

You turn in your answers. A few minutes later, a computer prints out your individual personality profile:

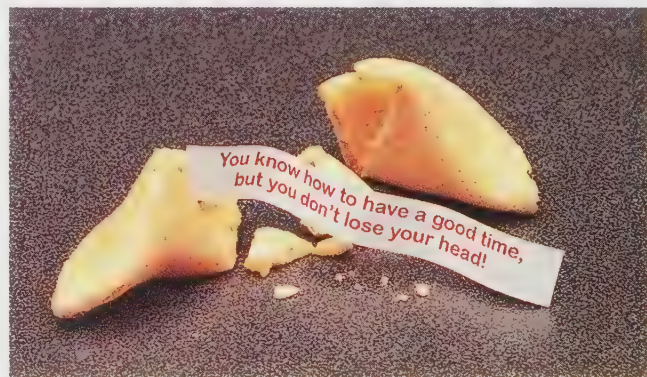
You have a need for other people to like and admire you, and yet you tend to be critical of yourself. While you have some

personality weaknesses, you are generally able to compensate for them. You have considerable unused capacity that you have not turned to your advantage. Disciplined and self-controlled on the outside, you tend to be worrisome and insecure on the inside. At times, you have serious doubts as to whether you have made the right decision or done the right thing. You prefer a certain amount of change and variety and become dissatisfied when hemmed in by restrictions and limitations. You also pride yourself as an independent thinker and do not accept others' statements without satisfactory proof. But you have found it unwise to be too frank in revealing yourself to others. At times you are extraverted, affable, and sociable, while at other times you are introverted, wary, and reserved. Some of your aspirations tend to be rather unrealistic. (Forer, 1949, p. 120)

Do you agree with this assessment?

Several experiments have been conducted along these lines with psychology classes (Forer, 1949; Marks & Kammann, 1980; Ulrich, Stachnik, & Stainton, 1963). Students started by filling out a questionnaire—one that looked fairly reasonable, not something as preposterous as the QUAZY. Several days later, each student received a sealed envelope with his or her name on it. Inside was a "personality profile," supposedly based on the student's answers to the questionnaire. The students were asked, "How accurately does this profile describe you?" About 90% rated it as good or excellent. Some expressed amazement at its accuracy: "I didn't realize until now that psychology was an exact science." None of them realized that everyone had received exactly the same personality profile—the same one you just read.

The students accepted this personality profile partly because it vaguely describes almost everyone, much like newspaper horoscopes do, and partly because people tend to accept almost *any* statement that an "expert" makes about them. Richard Kammann repeated this experiment but substituted a strange, unflattering personality profile that included statements like "your boundless energy is a little wearisome to your friends" and "you seem to find it impossible to work out a satisfactory adjustment to your problems."



People tend to accept almost any personality assessment that someone offers them, especially if it is stated in vague, general terms that each person can interpret to fit himself or herself. To determine the accuracy of a personality test, we need objective information, not just self-reports by satisfied customers.

More than 20% of the students rated this unlikely assortment of statements a “good to excellent” description of their own personality (Marks & Kammann, 1980).

The moral of the story is this: Psychological testing is tricky. If we want to know whether a particular test measures a particular person’s personality, we cannot simply ask whether that person thinks it does. Even if a test is totally worthless, many people will describe its results as a “highly accurate” description of themselves. To devise a psychological test that not only *appears* to work but also actually *does* work, we need to follow some elaborate procedures to design the test carefully and to determine its reliability and validity.

Standardized Personality Tests

Psychologists have devised a great variety of personality tests; they add new ones every year. A **standardized test** is *one that is administered according to specified rules and whose scores are interpreted in a prescribed fashion*. One important step for standardizing a test is to determine the distribution of scores for a large number of people. We need to know the mean score and the range of scores for people in general and the mean and the range for various special populations, such as severely depressed people. Given such information, we can determine whether a given individual’s score on the test is within the normal range or whether it is more typical of people with a particular disorder.

Most of the tests published in popular magazines have never been standardized. A magazine may herald an article: “Test Yourself: How Good Is Your Marriage?” or “Test Yourself: How Well Do You Control the Stress in Your Life?” After you take the test and compare your answers to the scoring key, the article may tell you that “if your score is greater than 80, you are doing very well . . . if it is below 20, you need to work on improving yourself!”—or some such nonsense. Unless the magazine states otherwise, you can safely assume that the author pulled the scoring norms out of thin air and never even bothered to make sure that the test items were clear and unambiguous.

Over the years, psychologists have developed an enormous variety of tests to measure both normal and abnormal personality. A great deal of research focused on trying to standardize their interpretation and measure their reliability and validity. We shall examine a few prominent examples and explore some creative possibilities for future personality measurement.

Objective Tests of Personality

Some of the most widely used personality tests are based on simple pencil-and-paper responses. We shall consider one of them—the MMPI—in some detail because it is the most

widely used of all personality tests (Piotrowski & Keller, 1989). We shall then examine the 16-PF more briefly.

The Minnesota Multiphasic Personality Inventory

The **Minnesota Multiphasic Personality Inventory** (mercifully abbreviated **MMPI**) consists of *a series of true-false questions intended to measure certain personality dimensions and clinical conditions such as depression*. The original MMPI, developed in the 1940s and still in use, has 550 items; *the second edition, MMPI-2*, published in 1990, has 567. Typical items are “my mother never loved me” and “I think I would like the work of a pharmacist.” (The items I mention in this text are rewordings of the actual items.)

The original MMPI was devised *empirically*—that is, by trial and error (Hathaway & McKinley, 1940). The authors developed hundreds of true-false questions that they thought might be useful for identifying personality dimensions. They put these questions to people who were known to be suffering from depression, paranoia, and other psychological disorders and to a group of hospital visitors, who were assumed to be psychologically normal. The researchers selected those items that most of the people in a given clinical group answered differently from most of the normal people. Their assumption was that, if you answer many questions just as depressed people usually answer them, you are probably depressed too. The MMPI had 10 scales—for reporting a depression score, a paranoia score, a schizophrenia score, and others. Later, other researchers found that they could use MMPI items to measure other dimensions of personality as well (Helmes & Reddon, 1993).

The result was a test that worked, and still works, moderately well in practice. For example, most people with scores above a certain level on the depression scale are in fact depressed. Some of the items on the MMPI made sense theoretically; some did not. For example, some items on the depression scale asked about feelings of helplessness or worthlessness, which are an important part of depression. But two other items were “I attend religious services frequently” and “occasionally I tease animals.” If you answered *false* to either of those items, you would get one point on the depression scale! These items were included simply because more depressed people than nondepressed people answered *false* to these items. Why they did is not obvious. (Perhaps depressed people do not tease animals just because they do hardly anything just for fun.)

Revisions of the Test

The MMPI was standardized in the 1940s. As time passed, the meaning of certain items, or at least of certain answers, changed. For example, how would you respond to the following item?

I believe I am important.

T F

In the 1940s, fewer than 10% of all people marked this item *true*. At the time, the word “important” meant about the same thing as “famous,” and people who called themselves important were thought to have an inflated view of themselves. Today, we are more likely to say that every person is important.

What about this item?

I like to play drop the handkerchief. T F

Drop the handkerchief, a game similar to tag, dropped out of popularity in the 1950s. Most people born since then have never even heard of the game, much less played it.

To bring the MMPI up to date, a group of psychologists rephrased some of the items, eliminated some, and added new ones to deal with drug abuse, suicidal ideas, Type A personality, and other issues that did not concern psychologists in the 1940s (Butcher, Graham, Williams, & Ben-Porath, 1990). Then, they tried out the new MMPI-2 on 2,600 people selected to resemble the current mix of age, sex, race, and education in the United States. In other words, the psychologists restandardized the test. (Any test must be restandardized from time to time. You may recall from the discussion of IQ tests in Chapter 9 that certain items once considered difficult are now considered relatively easy.) These psychologists also developed a new form, the MMPI-A, intended for use with adolescents.

The MMPI-2 has 10 clinical scales, as shown in Table 13.4. The various scales have 32 to 78 items each, scattered throughout the test, rather than clustered. Most people get at least a few points on each scale; a score above a certain level indicates a probable difficulty. Figure 13.20 shows how MMPI-2 scores are plotted.

The Generalizability of the MMPI

Your personality is such an integral part of who you are. Is it really possible for one test to measure personality for all kinds of people? In particular, is the MMPI (or MMPI-2 or

MMPI-A) a fair measure of personality for people of different ethnic and cultural backgrounds?

This is a difficult question to answer. In general, the means and ranges on each scale are about the same for many ethnic groups (Negy, Leal-Puente, Trainor, & Carlson, 1997). A few small differences in scores do occur, but they could reflect either real differences in personality or differences in interpreting what certain questions mean. Consequently, psychologists use the same norms for all groups, but they are slightly more cautious about interpreting the scores of racial minorities, especially those people who are most impoverished and least educated (Gynther, 1989).

Detection of Deception on the MMPI and Elsewhere

Suppose you were taking the MMPI or another personality test and you wanted to make yourself look mentally healthier than you really are. Could you lie on the test? Yes. Could anyone catch you in your lies? Probably.

The designers of the MMPI and the MMPI-2 included in their test certain items designed to identify people who consistently lie (Woychyshyn, McElheran, & Romney, 1992). For example, consider the items “I like every person I have ever met” and “occasionally I get angry at someone.” If you answer *true* to the first question and *false* to the second, you are either a saint or a liar. The test authors, convinced that there are more liars than saints, would give you one point for each of these answers on a special “lie scale.” If you get too many points on the lie scale, a psychologist will distrust your answers to the other items. Strangely enough, some people lie on the test to try to make themselves look bad. The test has some special items to detect that kind of faking also.

A similar method is used to detect deception on other types of tests. For example, many employers ask job applicants to fill out a questionnaire that asks them how much experience they have had with certain job-related skills.

TABLE 13.4 The Ten MMPI-2 Clinical Scales

SCALE	TYPICAL ITEM
Hypochondria (Hs)	I have chest pains several times a week. (T)
Depression (D)	I am glad that I am alive. (F)
Hysteria (Hy)	My heart frequently pounds so hard I can hear it. (T)
Psychopathic deviation (Pd)	I get a fair deal from most people. (F)
Masculinity-femininity (Mf)	I like to arrange flowers. (T = female)
Paranoia (Pa)	There are evil people trying to influence my mind. (T)
Psychasthenia (obsessive-compulsive) (Pt)	I save nearly everything I buy, even after I have no use for it. (T)
Schizophrenia (Sc)	I see, hear, and smell things that no one else knows about. (T)
Hypomania (Ma)	When things are dull I try to get some excitement started. (T)
Social introversion (Si)	I have the time of my life at parties. (F)

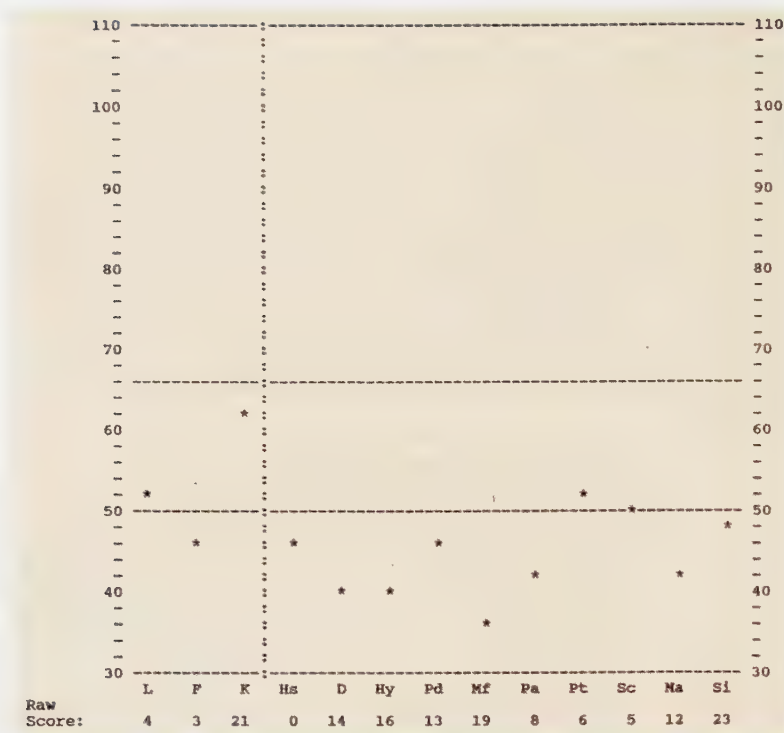


FIGURE 13.20 For the MMPI-2's 10 clinical scales, a score is plotted to profile an individual, as shown here. This is the profile of a middle-aged man with no psychological problems. A person with a disorder such as hypochondria or paranoia would have scores in the range of 65 or higher on the hypochondria or paranoia scales. (Source: Minnesota Multiphasic Personality Inventory-2, © by the Regents of the University of Minnesota. Data courtesy of R. J. Huber.)

What is to prevent eager applicants from exaggerating or even lying about their experience? To find out whether applicants are lying, some employers include among the authentic items a few bogus items referring to nonexistent tasks, as shown in Table 13.5.

According to the results of one study, almost half of all job applicants claimed to have experience with one or more nonexistent tasks (Anderson, Warner, & Spencer, 1984). Moreover, applicants who claimed a great deal of experience with nonexistent tasks also overstated their abilities on real tasks. An employer can use answers on bogus items as a correction factor. The more skill an applicant claims to have on a nonexistent task, the more the employer will discount that applicant's claims of skill on real tasks.

SOMETHING TO THINK ABOUT

Could you use this strategy in other situations? Suppose a political candidate promises to increase aid to college students. You are skeptical. How could you use the candidate's statements on other issues to help you decide whether to believe this promise? ★

Uses of the MMPI

The MMPI is useful to psychologists who want to measure personality for research purposes. It is also useful to clinical psychologists who want to learn something about a client before beginning therapy or who want an independent measure of how much a client's personality has changed during the course of therapy (McReynolds, 1985).

How informative are the results to the client who actually takes the MMPI test? In some cases, the results point out a problem to which the person had paid little attention. In other cases, however, the results do little more than restate the obvious. For example, suppose you gave the following answers:

I doubt that I will ever be successful.	True
I am glad that I am alive.	False
I have thoughts about suicide.	True
I am helpless to control the important events in my life.	True

A psychologist analyzes your answer sheet and tells you, "Your results show indications of depression." Yes, of course; you already knew that. But even in a case like this, the results can be useful—not just for telling you that you are depressed (which you already knew), but for measuring *how* depressed you are at this moment (a basis for comparison of future results).

The 16-PF Test

The **16-PF Test** is another widely used standardized personality test. The term "PF" stands for personality factors. The test *measures 16 factors, or traits, of personality*. Unlike the MMPI, which was intended primarily to identify abnormal personalities, the 16-PF Test was devised to assess various aspects of normal personality. Raymond Cattell (1965) used factor analysis to identify the traits that contribute most significantly to personality. As we saw earlier in this chapter, other psychologists used factor analysis to identify 5 major traits; Cattell found 35 and then narrowed them

TABLE 13.5 Part of an Employment Application Designed to Determine Whether Applicants Are Lying About Their Skills

How much experience have you had at ...	None	A Little	Much
Matrixing solvency files?	☐	☐	☐
Typing from audio-Fortran reports?	☐	☐	☐
Determining myopic weights for periodic tables?	☐	☐	☐
Resolving disputes by isometric analysis?	☐	☐	☐
Stocking solubility product constants?	☐	☐	☐
Planning basic entropy programs?	☐	☐	☐
Operating a matriculation machine?	☐	☐	☐

to 16. He then devised a test to measure each of those traits. Because of the large number of factors, the results of his test apply to a rather wide range of behaviors (Krug, 1978).

When someone takes the 16-PF Test, the results are printed out as a **personality profile**, a representation of that person's relative status on each dimension, as Figure 13.21 shows. By examining such a profile, a psychologist can determine the person's dominant personality traits.

Although the 16-PF Test was originally designed to assess normal personality, it does enable clinicians to identify various abnormalities, such as schizophrenia, depression, and alcoholism. Each disorder is associated with a characteristic personality profile (Figure 13.22). As with any test, this test should be used cautiously, especially with people

from different cultural backgrounds. Psychologists have translated this test into other languages, but something is often lost in translation. One study found that Mexican-Americans taking the 16-PF in Spanish had substantially different personality profiles than did Mexican-Americans taking supposedly the same test in English (Whitworth & Perry, 1990).

Projective Techniques

The MMPI, the 16-PF, and similar personality tests are easy to score and easy to handle statistically, but they restrict how a person can respond to a question. To find out

FIGURE 13.21 Personality profiles on the 16-PF Test for airline pilots, creative artists, and writers. A personality profile shows whether people are high or low on a given trait. In this sample, writers were the most imaginative group. (Adapted from *Handbook for the Sixteen Personality Factors*, copyright 1970 by the Institute for Personality and Ability Testing, Inc. Reproduced by permission of the copyright owners.)

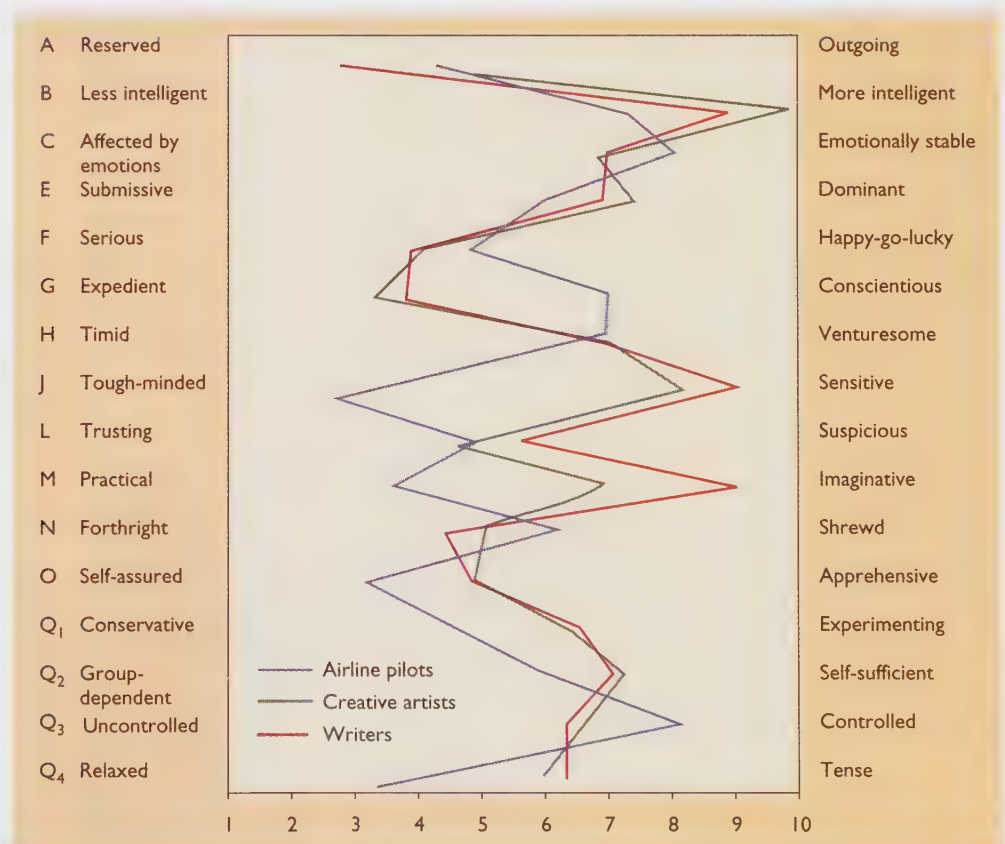




FIGURE 13.22 This personality profile, based on the 16-PF Test, shows that this person is high in guilt and low in “ego strength.” Cattell made up his own words for familiar concepts so that he could provide a precise definition that would not be confused with the everyday and vague meaning of a term such as *depression*. For example, *surgency* means something similar to cheerfulness and sociability. *Parmia* describes adventurousness or boldness. (From Cattell, 1965).

more, psychologists must ask open-ended questions that permit an unlimited range of responses.

To simply say “tell me about yourself” rarely evokes much information. In fact, most people find such invitations threatening. Many people are not fully honest even with themselves, much less with a psychologist they have just met.

Many people find it easier to discuss their problems in the abstract than in the first person. For instance, they might say, “I have a friend with this problem. Let me tell you my friend’s problem and ask what my friend should do.” They then describe their own problem. They are “projecting” their problem onto someone else, in Freud’s sense of the word—attributing their own characteristics to someone else.

Rather than discouraging projection, psychologists often make use of it. They use **projective techniques**, which are *designed to encourage people to project their personality characteristics onto ambiguous stimuli*. This strategy helps people to reveal themselves more fully than they normally would to a stranger, or even to themselves. Let’s consider two of the best-known projective techniques.

CONCEPT CHECK

7. Which of the following is a projective technique? **a.** A psychologist gives a child a set of puppets with instructions to act out a story about a family. **b.** A psycholo-

gist hands you a stack of cards, each containing one word, such as *tolerant*, with instructions to sort the cards into two piles—a stack of cards that apply to you and a stack of cards that do not apply to you. (Check your answer on page 513.)

The Rorschach Inkblots

The **Rorschach Inkblots**, a *projective test based on people’s interpretations of ten ambiguous inkblots*, is probably the most famous projective technique of personality. It was created by Hermann Rorschach, a Swiss psychiatrist, who was interested in art and the occult. He read a book of poems by Justinus Kerner, a mystic writer, who had made a series of random inkblots and had then written a poem about each one. Kerner believed that anything that happens at random reveals the influence of occult, supernatural forces.

Rorschach made his own inkblots but put them to a different use. He was familiar with a word-association test then in use where a person was given a word and was asked to say the first word that came to mind. Combining this approach with his inkblots, Rorschach showed people an inkblot and then asked them to say whatever came to mind (Pichot, 1984).

After testing a series of inkblots on his patients, Rorschach was impressed that their interpretations of the blots differed from his own. In a book published in 1921 (English translation 1942) he presented the 10 inkblots that

still constitute the Rorschach Inkblot Technique. (Originally, he had worked with a larger number, but the publisher insisted on cutting the number to 10 to save printing costs.) As other psychiatrists and psychologists began using these blots, they gradually developed the Rorschach into the projective technique we know today.

Administering the Rorschach The Rorschach Inkblot Technique consists of 10 cards similar to the one in Figure 13.23. Five are black and white; five are in color. A psychologist administering this procedure hands you a card and asks, “What might this be?” The instructions are intentionally vague. The assumption is that everything you do in an ill-defined situation will reveal something significant about your personality to the psychologist—so, the more poorly defined the situation, the better. The psychologist may keep a record of almost everything you do, including what you say you see, where and how you hold the cards, the length of any pauses between your responses, and so forth.

Sometimes people’s answers reveal much, either immediately or in response to a psychologist’s probes. Here is an example (Aronow, Reznikoff, & Moreland, 1995):

(Card 5). **Client:** Some kind of insect; it’s not pretty enough to be a butterfly.

Psychologist: Any association to that?

Client: It’s an ugly black butterfly, no colors.

Psychologist: What does that make you think of in your own life?

Client: You probably want me to say “myself.” Well, that’s probably how I thought of myself when I was younger—I never thought of myself as attractive—my sister was the attractive one. I was the ugly duckling—I did get more attractive as I got older.

CONCEPT CHECK

8. Why would it be impossible to receive a copy of the Rorschach Inkblot Test by mail, fill it out, and mail it back to a psychologist to evaluate your answers? (Check your answer on page 513.)



FIGURE 13.23 In the Rorschach Inkblot Test, people look at a pattern similar to this one and say what it looks like to them. The underlying theory is that, in an ambiguous situation, anything that someone does and says will reveal the individual’s personality.

Evaluation of the Rorschach Granted that people’s answers on the Rorschach often contain a wealth of personal information, the key issue is whether psychologists can accurately interpret that information. In the 1950s and 1960s, certain psychologists made exaggerated claims, even calling the Rorschach “an X ray of the mind.” Those claims provoked equally enthusiastic criticism. The trouble is partly that the Rorschach provides unstructured answers that must be interpreted by a psychologist who, in many cases, has preconceived notions about the client’s problems.

For example, one depressed man replied as follows to one blot: “It looks like a bat that has been squashed on the pavement under the heel of a giant’s boot” (Dawes, 1994, p. 149). Psychologist Robyn Dawes initially thought that this response illustrated the great power of the Rorschach: The client was expressing a sense of being overwhelmed and crushed by powers beyond his control. However, after further contemplation, Dawes realized that he had already known the client was depressed, so he interpreted the response accordingly. If a client with a history of violence had made the same response, the therapist would have focused on the aggressive nature of the giant’s foot stomp. If the client had been prone to hallucinations or to paranoia, the psychologist would have made still other interpretations. That is, Rorschach interpretations depend on the psychologist’s expectations at least as much as they do on what the client actually says. (Recall the need for double-blind procedures, as discussed in Chapter 2.)

A person’s responses to the Rorschach can be used in either of two ways, as a projective technique or as an objectively scored test. The more traditional method of using it as a projective technique focuses on the content, such as themes of violence or depression. Used in this manner, the Rorschach faces the problem just described: The interpretations depend heavily on the therapist’s expectations. Nevertheless, defenders of the technique reply that they use it only as a way of starting a conversation and getting clients to talk more freely about topics they might be reluctant to discuss (Aronow, Reznikoff, & Moreland, 1995). Used in that way, the Rorschach is beyond criticism, but its limitations are clear.

When the Rorschach is used as an objectively scored test, a psychologist counts particular kinds of responses and compares the results to the norms for the population. The use of standards developed by James Exner (1986) has greatly improved psychologists’ ability to agree on their interpretation of a person’s responses to the Rorschach. Thus, the test is now more reliable than it once was. Recall from Chapter 9, however, the difference between reliability and validity. Although interpretations of Rorschach responses are now reasonably reliable (repeatable), their validity is less certain.

Researchers have reported reasonably good results when using Rorschach responses to predict how well a client will respond in psychotherapy (Meyer & Handler,

1997). The test is also reasonably valid for identifying schizophrenia and for determining that someone has a psychological problem, without necessarily identifying the problem (Vincent & Harman, 1991). However, critics charge that, what the Rorschach does tell us, we could learn more easily or with greater certainty by other means (Wood, Nezowski, & Stejskal, 1996). In short, the Rorschach can be useful as a conversation starter and maybe as a supplement to other means of assessment, but it should not be the sole basis for making a confident judgment about someone. Many psychologists have serious qualms when their colleagues use Rorschach results to recommend that someone be committed to a mental hospital, or to recommend for or against a prisoner's parole (Dawes, 1994).

The Thematic Apperception Test

The **Thematic Apperception Test (TAT)** consists of pictures like the one shown in Figure 13.24. *The subject is asked to make up a story for each picture, describing what is happening, what events led up to the scene, and what will happen in the future.* The test was devised by Christiana Morgan and Henry Murray as a means of measuring people's needs; it was revised and published by Murray (1943) and later revised by others. There are 31 pictures in all; no more than 20 are used with a given individual. Some pictures are used just for men, women, boys, or girls, and some are used for anyone. The pictures are all somewhat ambiguous but, except for one (which is blank!), they provide a better-defined stimulus than does the Rorschach.

People who take the TAT are expected to identify with the people shown in the pictures. For this reason, men are given pictures showing mostly men, and women are given pictures showing mostly women. People usually tell stories that relate to recent events and concerns in their own lives, possibly including concerns that they would be reluctant to talk about openly.

For example, one young man told the following story about a picture of a man clinging to a rope:

This man is escaping. Several months ago he was beat up and shanghaied and taken aboard ship. Since then, he has been

mistreated and unhappy and has been looking for a way to escape. Now the ship is anchored near a tropical island and he is climbing down a rope to the water. He will get away successfully and swim to shore. When he gets there, he will be met by a group of beautiful native women with whom he will live the rest of his life in luxury and never tell anyone what happened. Sometimes he will feel that he should go back to his old life; but he will never do it. (Kimble & Garnezy, 1968, pp. 582–583)

This young man had entered divinity school, mainly to please his parents, but was quite unhappy there. He was wrestling with a secret desire to “escape” to a new life with greater worldly pleasures. In his story, he described someone doing what he really wanted to do but would not openly admit.

The TAT is often used in a clinical setting to induce clients to speak freely about their problems. It is also used for research purposes. As mentioned in Chapter 11, an investigator might measure someone's “need for achievement” by counting all the stories that he or she tells about achievement. The same could be done for aggression, passivity, control of outside events, or dominance. The investigator could then use these findings to study the forces that strengthen or weaken various needs.

When different psychologists examine the same TAT replies, they generally agree, more or less, on their interpretations. Most studies find a correlation of about .85 between different raters' interpretations—known as the *interrater reliability* (Cramer, 1996). When a given individual takes the TAT at two different times, months apart, the test-retest reliability is generally lower, usually less than .5 and sometimes much less than that (Cramer, 1996). That instability may not be a fault of the test, however; the test indicates people's needs and motivations, which do change over time.

Less Successful Projective Techniques

Based on the theory that your personality affects everything you do, some psychologists (and others) have tried analyzing people's handwriting. For example, perhaps people who dot their i's with a dash—*¿*—are especially energetic, or perhaps people who draw large loops above the line—as in *allow*—are highly idealistic. Carefully collected data, however, show only random relationships between handwriting and personality (Tett & Palmer, 1997).

Another projective technique is to offer children dolls and invite them to act out a story. The dolls might be chosen, for example, to represent two parents and a child. The idea is that a child who is unable or unwilling to describe physical or sexual abuse might act out a story in which the adults abuse the child. The problem is with the interpretation of the results: Suppose a child acts out sexual contact between two dolls. Should we surmise that an adult has sexually molested the child? Maybe not; the child may have discovered sex play with other children or may have watched late-night cable-television shows. This form of doll



FIGURE 13.24 In the Thematic Apperception Test, people look at a picture such as this one and tell a story about what is going on. Most people include material that relates to current concerns in their lives. Reprinted by permission of the publisher from Henry A. Murray, *Thematic Apperception Test*, Cambridge, Mass.: Harvard University Press, Copyright ©1943 by the President and Fellows of Harvard College, ©1971 by Henry A. Murray.